

Process variance and PIT over time, with and without bias correction

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Summary

This note follows the learned process variance and the one-step-ahead PIT year by year, to see whether the PIT flattens as the filter learns a smaller process variance. It is a companion to `claude-forecast-width-10.8.26`, which explains why the forecasts are too wide. It reaches four conclusions.

1. **With bias correction, the process variance falls steadily, by 20–50 times by 2017.** The median falls from 0.61 in 1991 to 0.01 (GEDI) and 0.03 (LandTrendr). Without correction it levels off at 0.1–0.25 after about 1995.
2. **The PIT flattens only a little, and only for GEDI.** For GEDI heteroskedastic R the PIT interquartile range (IQR) widens from 0.22 in 1995 to 0.28 by 2000 and then stays at 0.29–0.30 (calibrated: 0.5). For LandTrendr it does not widen at all (about 0.15–0.21), even though the process variance falls nine-fold over the same years.
3. **The bias correction centres the PIT; it does not flatten it.** Corrected, the mean and median PIT sit at about 0.54 in every year. Uncorrected they sit at about 0.27–0.36.
4. **The process variance stops mattering after the first few years.** The process share of the forecast variance is already small by the mid-1990s, so the remaining width comes from the observation error R , as shown in the forecast-width note.

1 Runs and definitions

- **Runs.** The two full grids, without the disturbance product (so no year is flagged), at all 538 sites, for each of the four observation-error models:
 - uncorrected: `experiments/fullRunNoBias`;
 - bias-corrected: `experiments/fullRunBias`, with `bias = list(rho = 0.9, denom = "N", skip = 1)`.

Year t runs from 1991 to 2017. Year 1990 is excluded because its forecast starts from the population-wide prior on the initial state.

- **Process variance** v_{t-1} . The posterior mean of the process variance after assimilating y_{t-1} :

$$v_{t-1} = \mathbb{E}[1/\tau \mid y_{1:t-1}] = \sum_g \pi_{t-1}(g)/\tau_g,$$

where $\pi_{t-1}(g)$ is the posterior weight of grid point τ_g (`hist$procVar` at $t-1$). This is the process variance used by the forecast of y_t , so it is plotted at year t , aligned with that year's PIT. (`claude-forecast-width-10.8.26` plots v_t , the same series one year later.)

- **PIT** u_t . The probability integral transform of y_t under its one-step-ahead predictive: the predictive probability of a value $\leq y_t$ (`hist$pit`). If the predictive is calibrated, u_t is uniform on $[0, 1]$: mean and median 0.5, quartiles 0.25 and 0.75, IQR 0.5. An IQR below 0.5 means the PIT is concentrated in the middle, i.e. the predictive is too wide. A median away from 0.5 means it is biased (below 0.5: forecasts too high).
- **Across-site summaries.** For each year, the median, IQR (25th–75th percentile) and mean over the 538 sites, of v_{t-1} and of u_t .
- **Coverage error of the central 50% interval.** The fraction of sites with $|2u_t - 1| \leq 0.5$ (the observation falls in the predictive's central 50% interval), minus 0.5. Positive values mean the forecast is too wide; 0 means calibrated.
- **Periods.** 1991–95, 1996–2000, 2001–05, 2006–11 and 2012–17, for the PIT histograms in Figure 2.

2 Results

model	year	v_{t-1}			PIT u_t			
		median	IQR	mean	median	IQR	IQR width	cov. err. 50%
GEDI het.	1991	0.607	0.583–0.653	0.627	0.40	0.32–0.50	0.18	0.36
	1995	0.043	0.034–0.057	0.122	0.58	0.46–0.68	0.22	0.28
	2000	0.014	0.010–0.032	0.091	0.52	0.40–0.67	0.28	0.27
	2010	0.010	0.005–0.035	0.070	0.54	0.40–0.69	0.29	0.22
	2017	0.012	0.004–0.071	0.083	0.56	0.40–0.70	0.30	0.22
GEDI const.	1995	0.105	0.098–0.116	0.147	0.56	0.47–0.65	0.18	0.36
	2000	0.032	0.030–0.040	0.084	0.51	0.42–0.62	0.20	0.36
	2017	0.014	0.008–0.040	0.073	0.55	0.45–0.65	0.21	0.31
LandTrendr het.	1995	0.229	0.216–0.262	0.257	0.56	0.44–0.65	0.21	0.41
	2000	0.086	0.078–0.101	0.116	0.50	0.43–0.61	0.18	0.44
	2017	0.026	0.020–0.047	0.074	0.53	0.46–0.61	0.15	0.37
LandTrendr const.	1995	0.255	0.238–0.288	0.280	0.56	0.44–0.65	0.21	0.41
	2000	0.099	0.090–0.117	0.128	0.51	0.42–0.60	0.18	0.44
	2017	0.029	0.023–0.052	0.082	0.53	0.46–0.61	0.15	0.37

Table 1: Bias-corrected runs, selected years: process variance v_{t-1} ($(\text{kg C m}^{-2})^2$) and PIT over the 538 sites. Calibrated PIT: median 0.5, IQR 0.25–0.75 (width 0.5), coverage error 0. Every year is in `fw_pit_time.csv`.

Process variance and PIT over time, no disturbance product, 538 sites

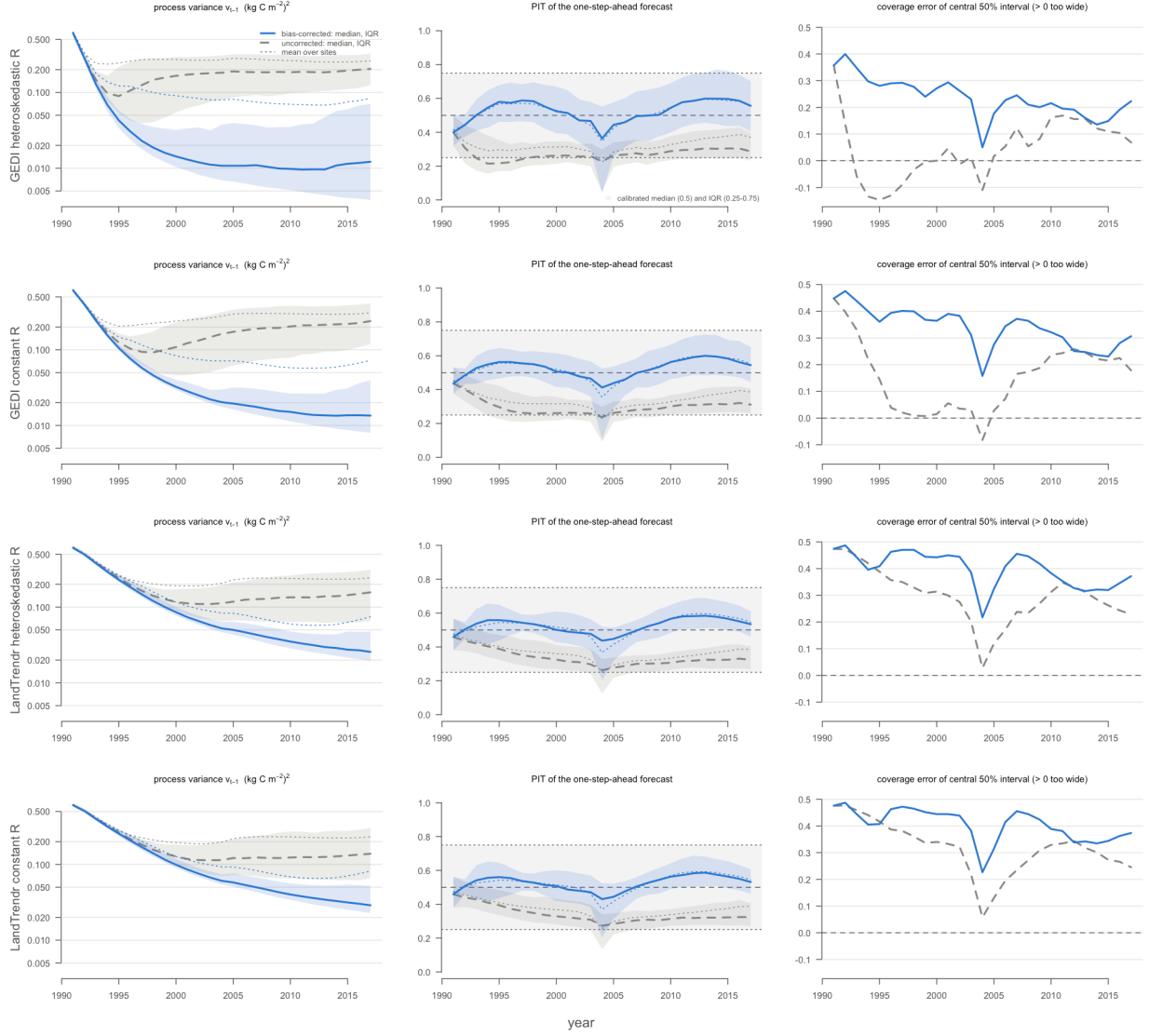


Figure 1: Per year, one row per observation-error model; blue = bias-corrected, gray dashed = uncorrected. *Left:* process variance v_{t-1} (log scale), median (thick line), IQR (band) and mean (dotted) over sites. *Middle:* PIT u_t , median, IQR and mean over sites. The shaded strip and dotted lines at 0.25 and 0.75 are the calibrated quantiles. *Right:* coverage error of the central 50% interval.

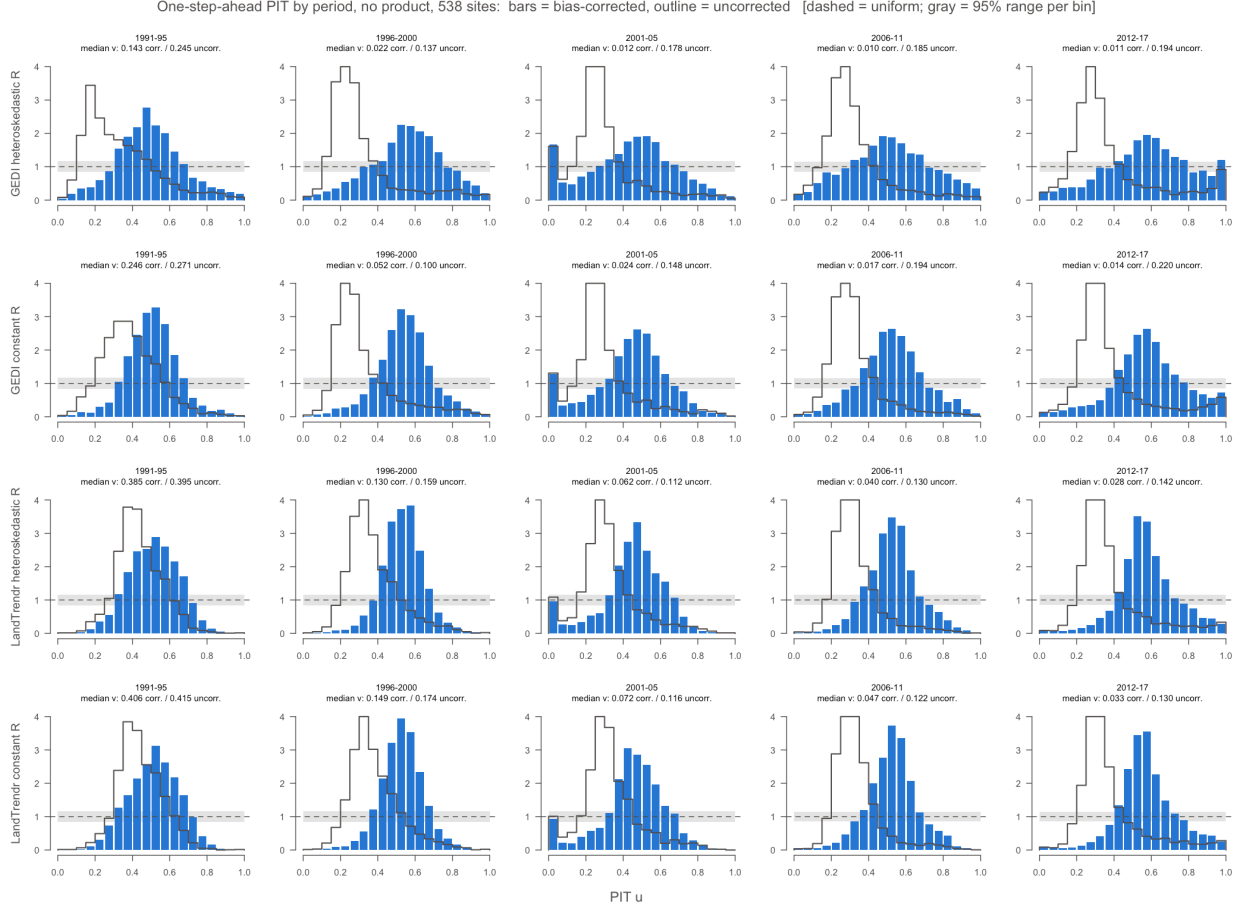


Figure 2: PIT histograms by period (all sites and years in the period). Bars: bias-corrected; outline: uncorrected; dashed: uniform; gray: 95% range of each bin under a uniform PIT. Panel titles give the median v_{t-1} over the period.

Process variance (Figure 1, left).

- **With correction it keeps falling.** The median falls fastest in the first five years and keeps falling slowly after that. For GEDI it reaches about 0.01, for LandTrendr about 0.03 by 2017.
- **Without correction it levels off.** After about 1995 the uncorrected median stays at 0.1–0.25, and for GEDI it creeps up: the filter uses process variance to absorb the forecast bias (as in the forecast-width note).
- **The mean is much larger than the median and levels off at 0.06–0.13 after 2000.** The distribution over sites is right-skewed: most sites learn a small process variance, and a minority with large year-to-year jumps learn a much larger one. For GEDI heteroskedastic R the upper quartile rises again after about 2013 (0.035 to 0.071), from sites whose biomass changes quickly late in the record.

PIT (Figure 1, middle and right; Figure 2).

- **Centring: the correction works in every year.** Averaged over 1996–2017 (excluding 2003–04, see below), the corrected mean PIT is 0.54 for every model. Uncorrected it is 0.33–0.36, with median 0.27–0.32: the uncorrected forecasts are too high throughout. The remaining 0.04 above 0.5 is the low disturbance tail of the predictive (forecast-width note, Section 3).
- **Concentration: the PIT flattens only a little, and only for GEDI.** For GEDI heteroskedastic R , the PIT IQR widens from 0.22 (1995) to 0.28 (2000) and then stays at 0.29–0.30, while the median v_{t-1} falls only from 0.014 to 0.010. The central 50% interval goes from 28 to 22 points too wide. For GEDI constant R the IQR stays at 0.18–0.21. For LandTrendr it *narrows* slightly (0.21 to 0.15) even though v_{t-1} falls nine-fold: the forecast errors shrink over time faster than the forecast variance, which stays dominated by R . In Figure 2 the corrected histograms stay humped in every period.
- **The uncorrected PIT can look better, by accident.** The uncorrected 50% coverage error is near 0 or even negative for GEDI in 1995–2005. That is not calibration: the bias pushes the observations away from the centre of a predictive that is too wide, and the two errors partly cancel. The uncorrected PIT IQR is in fact narrower (0.13–0.15) than the corrected one.
- **2003–04 is a disturbance year.** Many sites lose biomass in 2003–04, which the undisturbed forecast does not expect. The PIT drops (median 0.36–0.44 corrected), its lower quartile falls to 0.04–0.20, and the coverage error dips. This is real disturbance, not better calibration, so it is excluded from the averages above.
- **A late tail toward high PIT.** From about 2010 the corrected GEDI histograms grow a tail toward $u_t = 1$, i.e. forecasts that are too low at some sites. This matches the regrowth over-correction seen at site 300: b is learned in mature stands and is too negative for regrowing stands.

Interpretation. The process variance is the only part of the predictive variance that the filter learns, and it does shrink. But after the first few years it is a small part of the predictive variance (a median share of 1–4% of the full predictive variance from 1996 on, against 17–83% for R ; see

`fw_pit_time.csv` and the forecast-width note). Once it is small, shrinking it further cannot flatten the PIT. In the R-scaling experiment (`rs_pit.png`), reducing R flattens the PIT. That points to the observation-error model, not the process variance, as the thing to change.

Reproducing

Rscript `claude-working-docs/forecast_width_analysis.R` (section 6), from the repository root, with `EXP_DIR` set if the experiment folders are elsewhere. It writes `fw_pit_time.png`, `fw_pit_periods.png` and the per-year table `fw_pit_time.csv`.